Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Let α and β are two positive roots of $x^2 - 2ax + ab = 0$ where $0 < b < a$, then the value of $S_n = 1 + 2\frac{b}{a} + 3\frac{b}{a}^2 + \ldots\ldots + n\frac{b}{a}^{n-1}, \quad \forall n \in N$

cannot exceed

(A) $\frac{\alpha}{\beta}$

(B) $\left|\frac{\alpha+\beta}{\alpha-\beta}\right|$

(C) $\left|\frac{\beta}{\alpha}\right|$

(D) $\left(\frac{\alpha+\beta}{\alpha-\beta}\right)^4$

Q.2

The sum of the infinite series $\frac{1}{3} + \frac{3}{3.7} + \frac{5}{3.7.11} + \frac{7}{3.7.11.15} + \ldots\ldots\ldots$ is

(A) $\frac{1}{2}$

(B) $\frac{1}{3}$

(C) $\frac{1}{6}$

(D) $\frac{1}{4}$

Q.3

ABC is a right angled triangle in which $\angle B = 90^\circ$ and $BC = a$. If n points $L_1, L_2, \ldots, L_n$ on AB are such that AB is divided in $n+1$ equal parts and $L_1M_1, L_2M_2, \ldots, L_nM_n$ are line segments parallel to BC and $M_1, M_2, \ldots, M_n$ are on AC, then the sum of the lengths of $L_1M_1, L_2M_2, \ldots, L_nM_n$ is

(A) $\frac{a(n+1)}{2}$

(B) $\frac{a(n-1)}{2}$

(C) $\frac{an}{2}$

(D) None of these

Q.4

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The sum of $i-2-3i+4\ldots$ upto 100 terms, where $i=\sqrt{-1}$ is

(A) $50(1-i)$

(B) $25i$

(C) $25(1+i)$

(D) $100(1-i)$

Q.5

Sum of n terms of series $12+16+24+40+\ldots$ will be

(A) $2\left(2^n-1\right)+8n$

(B) $2\left(2^n-1\right)+6n$

(C) $3\left(2^n-1\right)+8n$

(D) $4\left(2^n-1\right)+8n$

Q.6

If m^{th} terms of the series $63+65+67+69+\ldots\ldots\ldots$ and $3+10+17+24+\ldots\ldots\ldots$ be equal, then $\mathrm{m}=$

(A) 11

(B) 12

(C) 13

(D) 15

Q.7

$\left(x+\frac{1}{x}\right)^2+\left(x^2+\frac{1}{x^2}\right)^2+$
$\left(\mathrm{x}^3+\frac{1}{\mathrm{x}^3}\right)^2$....upto n terms is

(A) $\frac{x^{2n}-1}{x^2-1} \times \frac{x^{2n+2}+1}{x^{2n}}+2n$

(B) $\frac{x^{2n}+1}{x^2+1} \times \frac{x^{2n+2}-1}{x^{2n}}-2n$

(C) $\frac{x^{2n}-1}{x^2-1} \times \frac{x^{2n}-1}{x^{2n}}-2n$

(D) None of these

Q.8

Let $\mathrm{S_n}(1 \leq \mathrm{n} \leq 9)$ denotes the sum of n terms of series $1+22+333+\ldots\ldots\ldots\ldots\ldots\ldots+$ 9999999999,
then for $2 \leq \mathrm{n} \leq 9$

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(A) $S_n - S_{n-1} = \frac{1}{9}\left(10^n - n^2 + n\right)$

(B) $S_n = \frac{1}{9}\left(10^n - n^2 + 2n - 2\right)$

(C) $9\left(S_n - S_{n-1}\right) = n\left(10^n - 1\right)$

(D) None of these

Q.9

If a^2, b^2, c^2 are in A.P. consider two statements
(i) $\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$ are in A.P.
(ii) $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in A.P.

(A) (i) and (ii) both correct

(B) (i) and (ii) both incorrect

(C) (i) correct (ii) incorrect

(D) (i) incorrect (ii) correct

Q.10

The value of the infinite product $6^{\frac{1}{2}} \times 6^{\frac{1}{2}} \times 6^{\frac{3}{8}} \times 6^{\frac{1}{4}} \times \ldots$ is

(A) 6

(B) 216

(C) 36

(D) ∞

Q.11

If $= S_n\left(1+3^{-1}\right)\left(1+3^{-2}\right)\left(1+3^{-4}\right)\left(1+3^{-8}\right)\ldots\left(1+3^{-2^n}\right)$
then S_∞ is equal to

(A) 1

(B) $\frac{1}{2}$

(C) $\frac{3}{2}$

(D) None

Q.12

What is the sum of the series
$1 + \frac{1}{8} + \frac{1.3}{8.16} + \frac{1.3.5}{8.16.24} + \ldots . \infty$?

(A) $\frac{2}{\sqrt{3}}$

(B) $2\sqrt{3}$

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(C) $\frac{\sqrt{3}}{2}$

(D) $\frac{1}{2\sqrt{3}}$

Q.13

The value of $0.0\overline{37}$ where $0.0\overline{37}$ stands for the number .0373737....., is :

(A) $37/1000$

(B) $37/990$

(C) $1/37$

(D) $1/27$

Q.14

$ABCD$ is a square of lengths $a, a \in N, a > 1$ Let $L_1, L_2, L_3, \ldots$ be points on BC such that $BL_1 = L_1L_2 = L_2L_3 = \ldots = 1$ and $M_1, M_2, M_3, \ldots$ be points on CD such that $\mathrm{CM_1 = M_1M_2 = M_2M_3 = \ldots = 1}$
Then, $\sum_{n=1}^{a-1}\left(AL_n^2 + L_nM_n^2\right)$ is equal to

(A) $\frac{1}{2}a(a-1)^2$

(B) $\frac{1}{2}a(a-1)(4a-1)$

(C) $\frac{1}{2}(a-1)(2a-1)(4a-1)$

(D) None of these

Q.15

If S_p denotes the sum of first n terms of an A.P. whose first term is a and $\frac{S_{nx}}{S_x}$ is independent of x, then $S_n =$

(A) P^3

(B) P^2a

(C) Pa^2

(D) a^3

Q.16

The sum of the series
$\frac{2}{3} + \frac{8}{9} + \frac{26}{27} + \frac{80}{81} + \ldots . n$ terms is:

(A) $n - \frac{1}{2}\left(3^n - 1\right)$

(B) $n + \frac{1}{2}\left(3^n - 1\right)$

(C) $n - \frac{1}{2}\left(1 + 3^{-n}\right)$

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(D) $n+\frac{1}{2}\left(3^{-n}-1\right)$

Q.17

The value of $\sum_{n=1}^{10} \sum_{m=1}^{10}\left(m^2+n^2\right)$ equals

(A) 4235

(B) 5050

(C) 7700

(D) None of these

Q.18

Concentric circles of radii $1, 2, 3, \ldots 100$ cm are drawn. The interior of the smallest circle is coloured red and the angular regions are coloured alternately green and red, so that no two adjacent regions are of the same colour. The total area of the green regions on sq cm is equal to

(A) 1000π

(B) 5050π

(C) 4950π

(D) 5151π

Q.19

The maximum sum of the series
$20+19\frac{1}{3}+18\frac{2}{3}+18+\ldots$ is

(A) 300

(B) 310

(C) $311\frac{2}{3}$

(D) $333\frac{1}{3}$

Q.20

What is the sum of the series $0.5+0.55+0.555+$ to n terms?

(A) $\frac{5}{9}\left[n-\frac{2}{9}\left(1-\frac{1}{10^n}\right)\right]$

(B) $\frac{1}{9}\left[5-\frac{2}{9}\left(1-\frac{1}{10^n}\right)\right]$

(C) $\frac{1}{9}\left[n-\frac{5}{9}\left(1-\frac{1}{10^n}\right)\right]$

(D) $\frac{5}{9}\left[n-\frac{1}{9}\left(1-\frac{1}{10^n}\right)\right]$

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Q.1 Answer:

$\frac{\alpha + \beta^4}{\alpha - \beta}$

Solution:

$S_n < S_\infty$

$\Rightarrow S_n < 1 + 2\frac{b}{a} + 3\frac{b}{a}^2 + \ldots .$ **upto** ∞

$\Rightarrow S_n < \frac{1}{1 - \frac{b}{a}} + \frac{1 \times \frac{b}{a}}{1 - \frac{b}{a}^2} = \frac{1}{1 - \frac{b}{a}^2} = \frac{a}{a - b}^2$

$\Rightarrow S_n < \frac{4a^2}{4a^2 - 4ab}^2$

Now, $\alpha + \beta = 2a$ **and** $\alpha\beta = ab$

$\Rightarrow S_n < \frac{\alpha + \beta^2}{\alpha + \beta^2 - 4\alpha\beta}^2 = \frac{\alpha + \beta^2}{\alpha - \beta^2}^2$

$\Rightarrow S_n < \frac{\alpha + \beta}{\alpha - \beta}^4$

Q.2 Answer:

$\frac{1}{2}$

Solution:

The nth **term of the series is**

$t_n = \frac{2n - 1}{3.7.11. - - - 4n - 1} = \frac{1}{2}\frac{4n - 1 - 1}{3.7.11. - - - 4n - 1}$

$= \frac{1}{2}\frac{1}{3.7.11. - - - 4n - 5} - \frac{1}{3.7.11. - - - 4n - 1}$

$= \frac{1}{2}u_{n-1} - u_n, \quad n \geq 2$

Where, $u_n = \frac{1}{3.7.11. - - - 4n - 1}$

$\Rightarrow S_n = t_1 + \sum_{k=2}^{n} t_k = \frac{1}{3} + \frac{1}{2}u_1 - u_2 + u_2 - u_3 - - - + u_{n-1} - u_n$

$= \frac{1}{3} + \frac{1}{2}u_1 - u_n = \frac{1}{3} + \frac{1}{2}\frac{1}{3} - u_n$

$\Rightarrow \lim_{n \to \infty} S_n = \frac{1}{3} + \frac{1}{6} - \frac{1}{2}\lim_{n \to \infty} u_n = \frac{1}{2} - 0 = \frac{1}{2}$

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Q.3 Answer:

$\frac{an}{2}$

Solution:

$\frac{AL_1}{AB} = \frac{L_1M_1}{BC}$
$\therefore \frac{1}{n+1} = \frac{L_1M_1}{a}$
$\therefore \quad L_1M_1 = \frac{a}{n+1}$
$\frac{AL_2}{AB} = \frac{L_2M_2}{BC}$
$\therefore \quad \frac{2}{n+1} = \frac{L_2M_2}{a}$
$\therefore L_2M_2 = \frac{2a}{n+1}$, **etc.**
- The required sum $= \frac{a}{n+1} + \frac{2a}{n+1} + \frac{3a}{n+1} +$
$\ldots + \frac{na}{n+1}$
$= \frac{a}{n+1}(1 + 2 + \ldots + n) = \frac{a}{n+1} \cdot \frac{n(n+1)}{2} = \frac{an}{2}$

Q.4 Answer:

$50(1 - i)$

Solution:

Let $S = i - 2 - 3i + 4 + 5i \ldots 100$ **terms** $S = i + 2i^2 + 3i^3 + 4i^4 + 5i^5 \ldots + 100i^{100}$
$\Rightarrow iS = i^2 + 2i^3 + 3i^4 \ldots + 99i^{100} + 100i^{101}$
$\Rightarrow \quad S - iS = i + i^2 + i^3 + i^4 + \ldots + i^{100} - 100i^{101}$
$\Rightarrow S(1 - i) = \frac{i\left(1 - i^{100}\right)}{1 - i} - 100i^{101}$
$\Rightarrow \quad S(1 - i) = -100i$
$\Rightarrow \quad S = \frac{-100i}{1 - i} = -50i(1 + i) = -50(i - 1)$
$= 50(1 - i)$

Q.5 Answer:

$4\left(2^n - 1\right) + 8n$

Solution:

Step-1: Reforming the given series

Let S **be the sum of the first** n **terms of the given series** $12 + 16 + 24 + 40 + \cdots$

i.e. $S = 12 + 16 + 24 + 40 + \cdots + t_n$, **where** t_n **is the** n^{th} **term of the series. To find** t_n **we shall rewrite the series as follows :**

$12 + 16 + 24 + 40 + \cdots$
$= \left(8 + 2^2\right) + \left(8 + 2^3\right) + \left(8 + 2^4\right) + \left(8 + 2^5\right) + \cdots$

So, using the above pattern, we get $t_n = 8 + 2^n$.

Step-2: Finding the value of S

We have

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$$
\begin{aligned}
S &= \left(8+2^2\right)+\left(8+2^3\right)+\left(8+2^4\right)+\left(8+2^5\right)+\cdots+\left(8+2^n\right) \\
&= (8+8+\cdots+8)+\left(2^2+2^3+\cdots+2^n\right) \\
&= 8n+\frac{2^2\left(2^{m-1}-1\right)}{2-1} \\
&= 8n+4\left(2^n-1\right)
\end{aligned}
$$

Here, we have used the fact that the sum of the first n terms a geometric series with the first term a and common ratio r is $\frac{a(r^n-1)}{r-1}$.

Q.6 Answer:

13

Solution:

Given series $63+65+67+69\ldots\ldots\ldots\ldots$
and $3+10+17+24+\ldots.$
Now from (i), m^{th} **term** $=(2m+61)$ **and** m^{th} **term of (ii) series** $=(7m-4)$ **Under condition**
$\Rightarrow 7m-4=2m+61 \Rightarrow 5m=65 \Rightarrow m=13.$

Q.7 Answer:

$\frac{x^{2n}-1}{x^2-1} \times \frac{x^{2n+2}+1}{x^{2n}}+2n$

Solution:

The series is $\left(x^2+x^4+x^6+\right.$ **up to** n **terms** $)$
$+\left(\frac{1}{x^2}+\frac{1}{x^4}+\frac{1}{x^6}+\right.$ **up to n terms** $)$
$+2\mathrm{n}=\frac{x^2\left(x^{2n}-1\right)}{x^2-1}+\frac{\frac{1}{x^2}\left(1-\frac{1}{x^{2n}}\right)}{1-\frac{1}{x^2}}+2n$
$=\frac{x^2\left(x^{2n}-1\right)}{x^2-1}+\frac{x^{2n}-1}{\left(x^2-1\right)x^{2n}}+2n$
$=\frac{x^{2n}-1}{x^2-1} \times \frac{x^{2n+2}+1}{x^{2n}}+2n$

Q.8 Answer:

$9\left(\mathrm{S_n}-\mathrm{S_{n-1}}\right)=\mathrm{n}\left(10^{\mathrm{n}}-1\right)$

Solution:

$\mathrm{S_n}=\frac{1}{9}(9)+\frac{2}{9}(99)+\frac{3}{9}(999)+\ldots.$
$=\frac{1}{9}\left[10+2.10^2+3.10^3+\ldots..\right]-\frac{1}{9}[1+2+3+\ldots.]$
$=\frac{1}{9}\,\mathrm{S}-\frac{1}{9}\frac{\mathrm{n}(\mathrm{n}+1)}{2}$
$\mathrm{S}=10+2.10^2+3.10^3+\ldots..+\mathrm{n}10^{\mathrm{n}}$
$10\,\mathrm{S}=10^2+2.10^3+\ldots\ldots+(\mathrm{n}-1)10^{\mathrm{n}}+\mathrm{n}\cdot 10^{\mathrm{n}+1}$
$-9\,\mathrm{s}=\left(10+10^2+\ldots.+10^{\mathrm{n}}\right)-\mathrm{n}10^{\mathrm{n}+1}$
$\mathrm{S}=\frac{\mathrm{n}}{9}10^{\mathrm{n}+1}-\frac{10^{\mathrm{n}+1}-10}{81}$
$\mathrm{S_n}=\frac{\mathrm{n}}{81}10^{\mathrm{n}+1}-\frac{10^{\mathrm{n}+1}-10}{9.81}-\frac{1}{9}\frac{\mathrm{n}(\mathrm{n}+1)}{2}$
$9S_n=\frac{(9n-1)10^{n+1}}{81}+\frac{10}{81}-\frac{n(n+1)}{2}$

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$\therefore \quad 9\left(S_n - S_{n-1}\right) = \frac{10^n}{81}$

$\{10(9n-1) - (9n-10)\} - n = n\left(10^n - 1\right)$

Q.9 Answer:

(i) and (ii) both correct

Solution:

Given $a^2,\ b^2 \cdot c^2$ **are in A.P.**

$\Rightarrow a^2 + (ab+bc+ca), b^2 + (ab+bc+ca)$

$c^2 + (ab + bc + ca)$ **are in A.P.**

$\Rightarrow (a+b)(a+c), (b+c)(b+a), \quad (c+a)(c+b)$

are in A.P.

$\Rightarrow \frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$ **are in A.P.**

[**Divide by** $(a+b)(b+c)(c+a)$]

Again, $a^2,\ b^2 \cdot c^2$ **are in A.P.**

$\Rightarrow \frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$ **are in A.P.**

$\Rightarrow \frac{a+b+c}{b+c}, \frac{a+b+c}{c+a}, \frac{a+b+c}{a+b}$ **are in A.P.**

$\Rightarrow \frac{a}{b+c} + 1, \frac{b}{c+a} + 1, \frac{c}{a+b} + 1$ **are in A.P.**

$\Rightarrow \frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ **are in A.P.**

Q.10 Answer:

36

Solution:

$X = 6^{(1/2+1/2+3/8+1/4+\ldots\ldots\ldots)}$

$= 6^{\left[\left(1\times\frac{1}{2}\right)+\left(2\times\frac{1}{4}\right)+\left(3\times\frac{1}{8}\right)+\left(4\times\frac{1}{16}\right)+\ldots..\right]}$

- It is arithmetic-geometric progression, $\therefore a = \frac{1}{2}; d = 1 \& r = \frac{1}{2}$

$\Rightarrow X = 6^2$

Q.11 Answer:

$\frac{3}{2}$

Solution:

$S_n = \left(1+3^{-1}\right)\left(1+3^{-2}\right)\left(1+3^{-4}\right)\left(1+3^{-4}\right)\ldots\left(1+3^{-2n}\right)$

$\Rightarrow \quad \left(1-3^{-1}\right)S_n$

$= \left(1-3^{-1}\right)\left(1+3^{-1}\right)\left(1+3^{-2}\right)\left(1+3^{-4}\right)\left(1+3^{-8}\right)\ldots\left(1+3^{-2^n}\right)$

$\Rightarrow \quad \frac{2}{3}S_n = \left(1-3^{-2}\right)\left(1+3^{-2}\right)\left(1+3^{-4}\right)\left(1+3^{-8}\right)$

$\ldots..\left(1+3^{-2n}\right)$

$= \left(1-3^{-4}\right)\left(1+3^{-4}\right)\left(1+3^{-8}\right)\ldots\left(1+3^{-2^n}\right)$

$= \left(1-3^{-8}\right)\left(1+3^{-8}\right)\ldots.\left(1+3^{-2^n}\right)$

$= \left(1-3^{-2^n}\right)\left(1+3^{-2^n}\right) = 1-\left(3^{-2^n}\right)^2 = 1-3^{-2^{n+1}}$

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$\Rightarrow \quad S_n = \frac{3}{2}\left(1 - 3^{-2^{n+1}}\right)$

$\therefore \quad S_\infty = \frac{3}{2}(1-0) = \frac{3}{2}$

Q.12 Answer:

$\frac{2}{\sqrt{3}}$

Solution:

As given the series is $S = 1 + \frac{1}{8} + \frac{1.3}{8.16} + \frac{1.3.5}{8.1.624} + \ldots . \infty$

On comparing this series with $S = (1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots \infty$

we get

$nx = \frac{1}{8}$ **...(i)**

and $\frac{n(n-1)}{2!}x^2 = \frac{1.3}{8.16}$

From Eqs. (1) and (2), we get $\frac{\frac{n(n-1)}{2!}x^2}{n^2x^2} = \frac{\frac{1.3}{8.16}}{\frac{1}{8}\cdot\frac{1}{8}}$

$\Rightarrow \frac{n-1}{2n} = \frac{3}{2}$

$\Rightarrow n - 1 = 3n$

$\Rightarrow \quad n = -\frac{1}{2}$

On putting this value in Eq. (i) $\Rightarrow \left(-\frac{1}{2}\right) x = \frac{1}{8}$

$\Rightarrow \quad x = -\frac{1}{4}$

But $S = (1+x)^n = \left(1 - \frac{1}{4}\right)^{-1/2}$

$= \left(\frac{3}{4}\right)^{-1/2} = \frac{2}{\sqrt{3}}.$

Q.13 Answer:

$37/990$

Solution:

The value of $0.0\overline{37}$ **stands for the number** $0.0373737 \ldots \ldots \ldots = 0.037 + 0.00037 + \ldots \ldots \ldots \ldots$

$$= \frac{37}{10^3} + \frac{37}{10^5} + \ldots \ldots \ldots \ldots =$$

$$\frac{37}{10^3}\left[1 + \frac{1}{100} + \ldots \ldots \ldots \ldots\right]$$

$$= \frac{37}{10^3}\left[\frac{1}{1 - \frac{1}{100}}\right] = \frac{37}{990}$$

Q.14 Answer:

$\frac{1}{2}a(a-1)(4a-1)$

Solution:

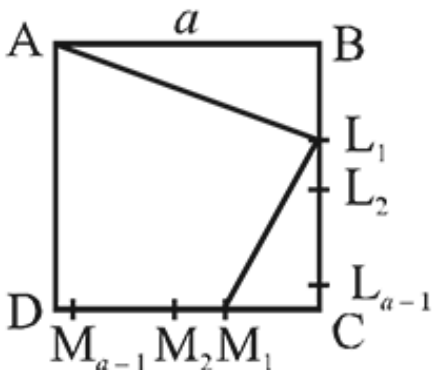
$AL_1^2 + L_1M_1^2 = \left(a^2 + 1^2\right) + \left\{(a-1)^2 + 1^2\right\}$

$AL_2^2 + L_2M_2^2 = \left(a^2 + 2^2\right) + \left\{(a-2)^2 + 2^2\right\}$

$AL_{a-1}^2 + L_{a-1}M_{a-1}^2 = a^2 + (a-1)^2 + \left\{1^2 + (a-1)^2\right\}$

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$\therefore$ **The required sum**



$= (a-1)a^2 + \left\{1^2 + 2^2 + \ldots + (a-1)^2\right\}$
$+2\left\{1^2 + 2^2 + \ldots + (a-1)^2\right\}$
$= (a-1)a^2 + 3 \cdot \frac{(a-1)a(2a-1)}{6}$
$= a(a-1)\left\{a + \frac{2a-1}{2}\right\} = \frac{a(a-1)(4a-1)}{2}$

Q.15 Answer:

P^2a

Solution:

$\frac{S_{nx}}{S_x} = \frac{\frac{nx}{2}[2a+(nx-1)d]}{\frac{x}{2}[2a+(x-1)d]}$
$= \frac{n[(2a-d)+nxd]}{(2a-d)+xd}$
For $\frac{S_{nx}}{S_x}$ **to be independent of** $x \quad 2a - d = 0 \therefore 2a = d$
now, $S_p = \frac{P}{2}[2a + (P-1)d] = P^2a$

Q.16 Answer:

$n + \frac{1}{2}\left(3^{-n} - 1\right)$

Solution:

Taking the sequence $3, 9, 27, 81, \ldots\ldots$ **Its** n^{th} **term** $= 3(3)^{n-1} = 3^n$
Also take the sequence $2, 8, 26, 80 \ldots\ldots$ **or** $(3-1), (9-1), (27-1), (81-1), \ldots$
Its nth **term** $= 3^n - 1$ **Hence,** n^{th} **term of the sequence**
$\frac{2}{3} + \frac{8}{9} + \frac{26}{27} + \frac{80}{81} + \ldots\ldots$ **is** $\frac{3^n-1}{3^n}$ **or** $1 - 3^{-n}$
Now the sum $\left(\mathrm{S}_n\right) = \Sigma\left(1 - 3^{-n}\right)$
$= n - \left(3^{-1} + 3^{-2} + \ldots + 3^{-n}\right)$
$= n - \frac{3^{-1}\left\{1-\left(3^{-1}\right)^n\right\}}{1-3^{-1}} = n - \frac{1}{2}\left(1 - 3^{-n}\right)$
$= n + \frac{1}{2}\left(3^{-n} - 1\right)$

Q.17 Answer:

7700

Solution:

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$$\sum_{n=1}^{10}\sum_{m=1}^{10}\left(m^2+n^2\right)$$

$$=\sum_{n=1}^{10}\left[\left(1^2+n^2\right)+\left(2^2+n^2\right)+\ldots+\left(10^2+n^2\right)\right]$$

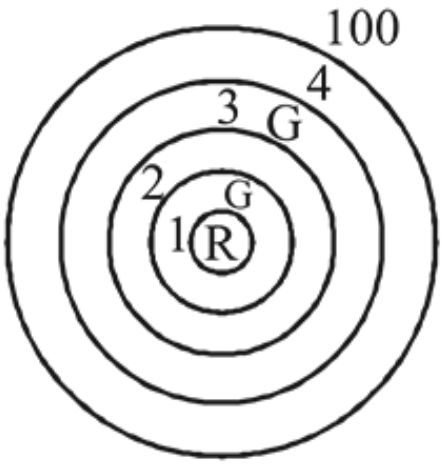
$$=10\left[(1)^2+(2)^2+\ldots+(10)^2\right]+10\left[(1)^2\right.$$
$$\left.+(2)^2+\ldots+(10)^2\right]$$
$$=\frac{20\cdot 10\cdot 11\cdot 21}{6}=7700$$

Q.18 Answer:

5050π

Solution:

$$(\pi\left[\left(r_2^2-r_1^2\right)+\left(r_4^2-r_3^2\right)\right.$$
$$\left.+\ldots+\left(r_{100}^2-r_{99}^2\right)\right]$$
$$\therefore \quad r_2-r_1=r_4-r_3=\ldots$$
$$=r_{100}-r_{99}=1$$
$$=\pi\left[r_1+r_2+r_3+r_4+\ldots+r_{100}\right]$$
$$=\pi[1+2+3+\ldots+100]$$
$$=5050\pi\text{sqcm}$$



Q.19 Answer:

310

Solution:

The given series is arithmetic whose first term $=20$, and common difference $=-\frac{2}{3}$ As the common difference is negative the terms will become negative after some stage. So the sum is maximum when all positive terms are added Now, for the positive terms $x_n \geq 0 \Rightarrow 20+(n-1)\times-\frac{2}{3}\geq 0$

$\Rightarrow 60-2(\mathrm{n}-1)\geq 0 \Rightarrow \mathrm{n}\leq 31$

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- The first 31 terms are non- negative ∴ Maximum sum

$= S_{31} = \frac{31}{2}\left[2 \times 20 + (31-1) \times -\frac{2}{3}\right] = 310$

Q.20 Answer:

$\frac{5}{9}\left[n - \frac{1}{9}\left(1 - \frac{1}{10^n}\right)\right]$

Solution:

Given $0.5 + 0.55 + 0.555 + \ldots$ **to** n

$= 5[0.1 + 0.11 + 0.111 + \ldots..$ **to** n **terms** $]$

$= \frac{5}{9}[0.9 + 0.99 + 0.999 + \ldots..$ **to** n **terms** $]$

$= \frac{5}{9}\left[\frac{9}{10} + \frac{99}{100} + \frac{999}{1000} + \ldots..\right.$ **to** n **terms** $]$

$= \frac{5}{9}\left[\begin{matrix}\left(1 - \frac{1}{10}\right) + \left(1 - \frac{1}{100}\right) + \left(1 - \frac{1}{1000}\right) + \ldots.. \\ \text{to n terms}\end{matrix}\right]$

$= \frac{5}{9}\left[\begin{matrix}\left(1 - \frac{1}{10}\right) + \left(1 - \frac{1}{10^2}\right) + \left(1 - \frac{1}{10^3}\right) + \ldots.. \\ \left(1 - \frac{1}{10^n}\right)\end{matrix}\right]$

$= \frac{5}{9}\left[n - \left(\frac{1}{10} + \frac{1}{10^2} + \ldots \cdot \frac{1}{10^n}\right)\right]$

$= \frac{5}{9}\left[n - \frac{1}{10}\frac{\left\{1 - \left(\frac{1}{10}\right)^n\right\}}{\left(1 - \frac{1}{10}\right)}\right]$

$= \frac{5}{9}\left[n - \frac{1}{9}\left(1 - \frac{1}{10^n}\right)\right]$

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